



ONE WATER OPERATING SYSTEM

The One Water AI Fieldbook

From first question to a governed utility pilot

PARTICIPANT WORKING EDITION 0.1 | 64-MODULE PORTFOLIO

Built for water, wastewater, stormwater, and One Water leadership

Powered by APAS.AI



THIS FIELDBOOK BELONGS TO

Your work. Your evidence. Your implementation record.

PARTICIPANT

Name, role, organization, and preferred contact information.

MY PROFESSIONAL OBJECTIVE

What do I need to understand, decide, build, improve, or prevent during the next six months?

MY SPONSOR AND PEER

Who supports this work, and who will challenge my assumptions honestly?



PRIVATE BY DESIGN

Do not put protected utility information into a course workbook.

Use generalized examples when a real record contains security-sensitive, personal, confidential, privileged, client, procurement, personnel, or operational information. The Fieldbook helps you reason about the work. It is not an approved repository for protected records.

MY CONFIDENTIALITY BOUNDARY

What information, systems, facilities, people, clients, security details, or records must I never enter or upload?

MY SAFE WORKING SUBSTITUTE

How will I generalize, redact, synthesize, or obtain approval for examples used in learning?



HOW TO USE THE FIELDBOOK

Every module leaves something useful behind.

01 | BEFORE

State what you believe and what you would decide before the lesson.

02 | LEARN

Record the relationship, mechanism, distinction, or consequence that matters.

03 | APPLY

Connect the idea to one real utility task, role, asset, record, or decision.

04 | KEEP

Add one durable record to your implementation portfolio.



STARTING POINT

Record the baseline before the curriculum changes your answers.

CURRENT FLUENCY

What can I explain today about models, prompts, retrieval, knowledge graphs, skills, agents, and governance?

CURRENT PRACTICE

Where is my organization already using, testing, buying, or discussing AI?

CURRENT CONCERN

What could go wrong technically, professionally, organizationally, or publicly?

CURRENT OPPORTUNITY

What work is slow, fragmented, repetitive, difficult to retrieve, or hard to defend?

AI Fluency for One Water

Build a shared language for AI and connect every term to a recognizable utility task, decision, or consequence.

THIS COURSE ADDS

- 01 AI use-case boundary card
- 02 AI family map
- 03 prompt and context anatomy sheet
- 04 multimodal utility opportunity card
- 05 system classifier
- 06 reusable utility prompt pattern
- 07 first utility Claude Skill specification
- 08 AI terms and decision field card



MODULE 01

AI and utility work

Separate useful assistance from accountable authority

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 01 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Separate useful assistance from accountable authority

KEEP

Add this record to the implementation portfolio: AI use-case boundary card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 02

How we arrived here

Understand rules, statistics, machine learning, deep learning, and generative AI as related but different approaches

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 02 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence. Learning job: Understand rules, statistics, machine learning, deep learning, and generative AI as related but different approaches

KEEP

Add this record to the implementation portfolio: AI family map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 03

Models, tokens, prompts, and context

Explain what a model receives and produces without treating it as a person

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 03 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Explain what a model receives and produces without treating it as a person

KEEP

Add this record to the implementation portfolio: prompt and context anatomy sheet.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 04

Language, images, audio, and multimodal AI

Recognize what changes when systems work across more than text

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 04 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Recognize what changes when systems work across more than text

KEEP

Add this record to the implementation portfolio: multimodal utility opportunity card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 05

Chatbot, retrieval, workflow, agent, or agentic system

Choose the least complicated architecture that can perform a task

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 05 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Choose the least complicated architecture that can perform a task

KEEP

Add this record to the implementation portfolio: system classifier.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 06

Working effectively with an AI assistant

Frame a task, provide context, define output, test assumptions, and revise

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 06 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Frame a task, provide context, define output, test assumptions, and revise

KEEP

Add this record to the implementation portfolio: reusable utility prompt pattern.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 07

Claude Skills and reusable organizational instructions

Understand a Skill as a versioned package of instructions, references, scripts, and assets that teaches a repeatable workflow

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 07 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Understand a Skill as a versioned package of instructions, references, scripts, and assets that teaches a repeatable workflow

KEEP

Add this record to the implementation portfolio: first utility Claude Skill specification.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 08

Fluency checkpoint

Explain a proposed AI use case to an operator, executive, technologist, and procurement professional

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 08 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Explain a proposed AI use case to an operator, executive, technologist, and procurement professional

KEEP

Add this record to the implementation portfolio: AI terms and decision field card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review

Data, Knowledge, and Trusted Context

Understand why useful AI depends on connected records, shared meaning, current sources, permission, and traceable evidence.

THIS COURSE ADDS

- 09 readiness diagnosis
- 10 data-quality control card
- 11 lineage trace
- 12 utility concept map
- 13 knowledge relationship map
- 14 retrieval comparison sheet
- 15 RAG answer contract
- 16 data and knowledge readiness map



MODULE 09

Data before AI

Diagnose why AI amplifies disconnected or poor-quality data

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 09 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Diagnose why AI amplifies disconnected or poor-quality data

KEEP

Add this record to the implementation portfolio: readiness diagnosis.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 10

Data quality and observability

Identify completeness, accuracy, timeliness, consistency, and monitoring needs

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 10 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Identify completeness, accuracy, timeliness, consistency, and monitoring needs

KEEP

Add this record to the implementation portfolio: data-quality control card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 11

Metadata, lineage, and provenance

Explain where a record came from, how it changed, and why it may be trusted

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 11 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Explain where a record came from, how it changed, and why it may be trusted

KEEP

Add this record to the implementation portfolio: lineage trace.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 12

Taxonomy and ontology

Build shared language and relationships without beginning with software

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 12 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Build shared language and relationships without beginning with software

KEEP

Add this record to the implementation portfolio: utility concept map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 13

Knowledge graphs and the systems lens

See assets, people, records, decisions, sources, and consequences as connected relationships

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 13 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: See assets, people, records, decisions, sources, and consequences as connected relationships

KEEP

Add this record to the implementation portfolio: knowledge relationship map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 14

Embeddings and vector retrieval

Explain similarity search, what it can retrieve, and what it cannot prove

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 14 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Explain similarity search, what it can retrieve, and what it cannot prove

KEEP

Add this record to the implementation portfolio: retrieval comparison sheet.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 15

Retrieval-augmented generation

Design a source-grounded answer path with citations and access boundaries

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 15 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Design a source-grounded answer path with citations and access boundaries

KEEP

Add this record to the implementation portfolio: RAG answer contract.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 16

Context engineering and the trusted answer

Select the minimum useful context for a task and preserve its source and limitation

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 16 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Select the minimum useful context for a task and preserve its source and limitation

KEEP

Add this record to the implementation portfolio: data and knowledge readiness map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review

Agents, Skills, and Orchestration

See how AI moves from a response to bounded work, then design the instructions, tools, handoffs, permissions, and stop conditions required to keep people in control.

THIS COURSE ADDS

- 17 architecture decision record
- 18 agent-loop trace
- 19 dependency and authority map
- 20 tool and connector register
- 21 working Claude Skill package
- 22 state and record plan
- 23 handoff and conflict contract
- 24 Utility Agent Canvas and agent experience map



MODULE 17

Before the agent

Compare a model response, chatbot, retrieval answer, workflow, and agent

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 17 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Compare a model response, chatbot, retrieval answer, workflow, and agent

KEEP

Add this record to the implementation portfolio: architecture decision record.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 18

Inside the agent loop

Follow goal, environment, state, plan, action, observation, evaluation, adjustment, and stop

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Follow goal, environment, state, plan, action, observation, evaluation, adjustment, and stop

KEEP

Add this record to the implementation portfolio: agent-loop trace.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 19

Agent anatomy

Examine model, instructions, environment, sources, tools, memory, identity, permissions, evaluation, and human owner

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 19 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Examine model, instructions, environment, sources, tools, memory, identity, permissions, evaluation, and human owner

KEEP

Add this record to the implementation portfolio: dependency and authority map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 20

Tools, connectors, and the Model Context Protocol

Understand how an agent reaches approved systems and why connection does not equal authority

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Understand how an agent reaches approved systems and why connection does not equal authority

KEEP

Add this record to the implementation portfolio: tool and connector register.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 21

Skills as organizational recipes

Design a reusable Skill with instructions, references, scripts, assets, triggers, and tests

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 21 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Design a reusable Skill with instructions, references, scripts, assets, triggers, and tests

KEEP

Add this record to the implementation portfolio: working Claude Skill package.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 22

Memory, state, and durable records

Separate temporary context, task state, institutional memory, and authoritative records

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 22 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Separate temporary context, task state, institutional memory, and authoritative records

KEEP

Add this record to the implementation portfolio: state and record plan.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 23

Orchestration and multi-agent handoffs

Coordinate specialist roles, shared state, communication, disagreement, evidence, failure, and human escalation

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 23 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Coordinate specialist roles, shared state, communication, disagreement, evidence, failure, and human escalation

KEEP

Add this record to the implementation portfolio: handoff and conflict contract.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 24

Design the smallest useful agent

Choose agenticness according to value, consequence, readiness, review, interaction burden, and recovery needs

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence. Learning job: Choose agenticness according to value, consequence, readiness, review, interaction burden, and recovery needs

KEEP

Add this record to the implementation portfolio: Utility Agent Canvas and agent experience map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review

Utility Applications and Opportunity Design

Move from generic AI excitement to a disciplined portfolio of opportunities grounded in utility work, available evidence, affected people, and measurable value.

THIS COURSE ADDS

- 25 operations opportunity card
- 26 maintenance opportunity card
- 27 evidence-chain card
- 28 customer-service use-case card
- 29 capital-delivery opportunity card
- 30 One Water relationship map
- 31 administrative opportunity card
- 32 prioritized utility opportunity portfolio



MODULE 25

Operations and operator support

Examine bounded advisory, retrieval, shift, and exception-management patterns

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 25 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Examine bounded advisory, retrieval, shift, and exception-management patterns

KEEP

Add this record to the implementation portfolio: operations opportunity card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 26

Maintenance and asset management

Connect work orders, history, inspection, condition, failure, and planning

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 26 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Connect work orders, history, inspection, condition, failure, and planning

KEEP

Add this record to the implementation portfolio: maintenance opportunity card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 27

Water quality, laboratory, and compliance support

Trace evidence, limits, reviews, and certification boundaries

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 27 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Trace evidence, limits, reviews, and certification boundaries

KEEP

Add this record to the implementation portfolio: evidence-chain card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 28

Customer service and public communication

Improve retrieval, consistency, routing, drafting, and accessibility while protecting privacy

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 28 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Improve retrieval, consistency, routing, drafting, and accessibility while protecting privacy

KEEP

Add this record to the implementation portfolio: customer-service use-case card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 29

Capital planning, engineering, and project delivery

Connect requirements, risks, changes, schedules, records, and decision history

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 29 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Connect requirements, risks, changes, schedules, records, and decision history

KEEP

Add this record to the implementation portfolio: capital-delivery opportunity card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 30

Stormwater, wastewater, drinking water, and One Water coordination

Examine cross-domain relationships without erasing domain-specific authority

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 30 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Examine cross-domain relationships without erasing domain-specific authority

KEEP

Add this record to the implementation portfolio: One Water relationship map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 31

Finance, procurement, legal, and administration

Identify high-volume knowledge work and its evidence and approval requirements

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 31 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Identify high-volume knowledge work and its evidence and approval requirements

KEEP

Add this record to the implementation portfolio: administrative opportunity card.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 32

Opportunity portfolio and prioritization

Compare value, feasibility, consequence, readiness, cost, and adoption effort

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 32 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Compare value, feasibility, consequence, readiness, cost, and adoption effort

KEEP

Add this record to the implementation portfolio: prioritized utility opportunity portfolio.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review

Governance, Security, and Human Authority

Build governance into the design so that the organization knows what the system may do, what it may not do, who checks it, and what evidence survives.

THIS COURSE ADDS

- 33 use-case risk tier
- 34 privacy and data-use record
- 35 threat and control map
- 36 permission matrix
- 37 authority, content-integrity, and escalation map
- 38 evaluation and observability plan
- 39 AI procurement scorecard
- 40 AI lifecycle governance record



MODULE 33

Risk and consequence tiers

Match review and control strength to possible harm and reversibility

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 33 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Match review and control strength to possible harm and reversibility

KEEP

Add this record to the implementation portfolio: use-case risk tier.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 34

Privacy, confidentiality, and data rights

Identify personal, sensitive, contractual, and restricted information boundaries

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 34 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Identify personal, sensitive, contractual, and restricted information boundaries

KEEP

Add this record to the implementation portfolio: privacy and data-use record.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 35

Cybersecurity and adversarial failure

Recognize prompt injection, unsafe tool use, data leakage, poisoned context, and excessive privilege

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 35 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Recognize prompt injection, unsafe tool use, data leakage, poisoned context, and excessive privilege

KEEP

Add this record to the implementation portfolio: threat and control map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 36

Identity, permissions, and segregation of duties

Separate who asks, retrieves, recommends, approves, writes, and administers

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 36 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Separate who asks, retrieves, recommends, approves, writes, and administers

KEEP

Add this record to the implementation portfolio: permission matrix.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 37

Human authority, escalation, content integrity, and stop conditions

Name the accountable owner, recognize manipulated content, and design pause, refusal, provenance, and escalation rules

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 37 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence. Learning job: Name the accountable owner, recognize manipulated content, and design pause, refusal, provenance, and escalation rules

KEEP

Add this record to the implementation portfolio: authority, content-integrity, and escalation map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 38

Evaluation, testing, observability, and continuous assurance

Test usefulness, accuracy, source coverage, refusal behavior, resilience, cost, drift, traces, and incident signals

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 38 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence. Learning job: Test usefulness, accuracy, source coverage, refusal behavior, resilience, cost, drift, traces, and incident signals

KEEP

Add this record to the implementation portfolio: evaluation and observability plan.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 39

Vendor, model, and procurement governance

Compare claims, architecture, data use, service boundaries, exit terms, and evidence

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 39 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Compare claims, architecture, data use, service boundaries, exit terms, and evidence

KEEP

Add this record to the implementation portfolio: AI procurement scorecard.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 40

The governed AI application record

Assemble purpose, sources, roles, controls, evaluations, release gates, monitoring, incidents, change history, retirement, and approvals

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 40 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Assemble purpose, sources, roles, controls, evaluations, release gates, monitoring, incidents, change history, retirement, and approvals

KEEP

Add this record to the implementation portfolio: AI lifecycle governance record.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review

Strategy, Economics, and Organizational Change

Turn isolated experiments into a sequenced organizational program with value measures, ownership, funding, workforce preparation, and an operating rhythm.

THIS COURSE ADDS

- 41 problem statement
- 42 value baseline and benefits register
- 43 total-cost and operating-stack model
- 44 sourcing decision record
- 45 operating-model canvas
- 46 adoption and workforce capability plan
- 47 portfolio roadmap
- 48 investment case and one-year roadmap



MODULE 41

Problem formulation before technology selection

Convert a complaint or idea into a measurable professional problem

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 41 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Convert a complaint or idea into a measurable professional problem

KEEP

Add this record to the implementation portfolio: problem statement.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 42

Value, baseline, and measurement

Define current effort, error, delay, risk, rework, service outcomes, benefit ownership, and measurement cadence

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 42 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence. Learning job: Define current effort, error, delay, risk, rework, service outcomes, benefit ownership, and measurement cadence

KEEP

Add this record to the implementation portfolio: value baseline and benefits register.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 43

The full cost and operating stack of AI

Account for models, infrastructure, integration, data, LLMOps, observability, review, security, change, and failure

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 43 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence. Learning job: Account for models, infrastructure, integration, data, LLMOps, observability, review, security, change, and failure

KEEP

Add this record to the implementation portfolio: total-cost and operating-stack model.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 44

Build, buy, partner, or do nothing

Compare strategic control, speed, capability, dependency, and exit cost

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 44 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Compare strategic control, speed, capability, dependency, and exit cost

KEEP

Add this record to the implementation portfolio: sourcing decision record.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 45

AI operating model and decision rights

Define sponsors, product owners, technical owners, stewards, assurance, users, and escalation

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 45 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Define sponsors, product owners, technical owners, stewards, assurance, users, and escalation

KEEP

Add this record to the implementation portfolio: operating-model canvas.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 46

Workforce capability and responsible adoption

Map changing tasks, participation, training, process redesign, communication, resistance, decision rights, and professional authority

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 46 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence. Learning job: Map changing tasks, participation, training, process redesign, communication, resistance, decision rights, and professional authority

KEEP

Add this record to the implementation portfolio: adoption and workforce capability plan.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 47

Portfolio governance and sequencing

Fund learning in stages and stop weak initiatives before they become sunk costs

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 47 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Fund learning in stages and stop weak initiatives before they become sunk costs

KEEP

Add this record to the implementation portfolio: portfolio roadmap.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 48

Executive narrative and one-year roadmap

Explain the problem, opportunity, evidence, risk, investment, and sequence

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 48 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Explain the problem, opportunity, evidence, risk, investment, and sequence

KEEP

Add this record to the implementation portfolio: investment case and one-year roadmap.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review

Applied AI Studios

Build enough to understand the mechanics, expose failure, and make better implementation decisions. No programming background is required.

THIS COURSE ADDS

- 49 tested prompt pattern
- 50 RAG test record
- 51 tested Claude Skill
- 52 tool-use evidence log
- 53 production-grade agent trace
- 54 multi-agent decision record
- 55 failure and recovery log
- 56 release rehearsal evidence package



MODULE 49

Prompt and context studio

Improve an unclear utility task through role, context, constraints, output, and evaluation

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 49 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Improve an unclear utility task through role, context, constraints, output, and evaluation

KEEP

Add this record to the implementation portfolio: tested prompt pattern.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 50

Source-grounded answer studio

Compare unsupported generation with governed retrieval and citations

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 50 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Compare unsupported generation with governed retrieval and citations

KEEP

Add this record to the implementation portfolio: RAG test record.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 51

Claude Skill studio

Build, trigger, test, revise, and document a reusable utility workflow

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 51 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Build, trigger, test, revise, and document a reusable utility workflow

KEEP

Add this record to the implementation portfolio: tested Claude Skill.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 52

Tool-use and permission studio

Connect a bounded read-only tool and inspect every request and result

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 52 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Connect a bounded read-only tool and inspect every request and result

KEEP

Add this record to the implementation portfolio: tool-use evidence log.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 53

Agent workflow studio

Build a goal-to-result loop with a defined environment, tool policy, evaluation, retry limits, observability, and escalation

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 53 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Build a goal-to-result loop with a defined environment, tool policy, evaluation, retry limits, observability, and escalation

KEEP

Add this record to the implementation portfolio: production-grade agent trace.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 54

Multi-agent decision-room studio

Observe specialist perspectives, shared evidence, disagreement, synthesis, and human decision

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 54 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Observe specialist perspectives, shared evidence, disagreement, synthesis, and human decision

KEEP

Add this record to the implementation portfolio: multi-agent decision record.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 55

Failure and red-team studio

Trigger unsupported claims, stale sources, instruction attacks, missing authority, and endless retries

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 55 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Trigger unsupported claims, stale sources, instruction attacks, missing authority, and endless retries

KEEP

Add this record to the implementation portfolio: failure and recovery log.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 56

Evaluation and pilot rehearsal

Test historical cases, record metrics, inspect human effort, exercise failure alerts and rollback, and prepare a controlled demonstration

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 56 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Test historical cases, record metrics, inspect human effort, exercise failure alerts and rollback, and prepare a controlled demonstration

KEEP

Add this record to the implementation portfolio: release rehearsal evidence package.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review

Capstone, From Opportunity to Governed Pilot

Convert one real organizational opportunity into a pilot that leadership can understand, challenge, measure, govern, and stop when necessary.

THIS COURSE ADDS

- 57 capstone charter
- 58 current-state map
- 59 readiness plan
- 60 architecture decision record
- 61 pilot control plan
- 62 measurement and financial plan
- 63 90-day pilot brief and one-year roadmap
- 64 defended implementation portfolio



MODULE 57

Capstone charter and stakeholder agreement

Confirm the problem, sponsor, affected people, scope, and decision process

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 57 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Confirm the problem, sponsor, affected people, scope, and decision process

KEEP

Add this record to the implementation portfolio: capstone charter.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 58

Current-state process and baseline

Document how the work happens today and where cost, delay, risk, or rework occurs

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 58 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Document how the work happens today and where cost, delay, risk, or rework occurs

KEEP

Add this record to the implementation portfolio: current-state map.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 59

Data and knowledge readiness

Identify approved sources, missing records, definitions, permissions, and provenance

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 59 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Identify approved sources, missing records, definitions, permissions, and provenance

KEEP

Add this record to the implementation portfolio: readiness plan.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 60

Architecture and workflow decision

Select the least complicated technical pattern and justify every component

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 60 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Select the least complicated technical pattern and justify every component

KEEP

Add this record to the implementation portfolio: architecture decision record.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 61

Governance, security, and human authority

Complete the controls, permissions, review, incident, and escalation design

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 61 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Complete the controls, permissions, review, incident, and escalation design

KEEP

Add this record to the implementation portfolio: pilot control plan.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 62

Evaluation, economics, and success criteria

Define test cases, baseline, thresholds, cost limits, review effort, and stop conditions

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 62 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Define test cases, baseline, thresholds, cost limits, review effort, and stop conditions

KEEP

Add this record to the implementation portfolio: measurement and financial plan.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 63

Ninety-day pilot and one-year adoption roadmap

Sequence discovery, build, test, review, decision, and possible expansion

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 63 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Sequence discovery, build, test, review, decision, and possible expansion

KEEP

Add this record to the implementation portfolio: 90-day pilot brief and one-year roadmap.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



MODULE 64

Capstone defense and professional commitment

Present the case, answer challenges, revise the plan, and state the next accountable action

BEFORE

What do I believe now? What decision would I make before the lesson, and what evidence would I use?

LEARN

What relationship, mechanism, distinction, or consequence changed my understanding?

QUESTIONS

What remains unclear? Which claim, source, example, or assumption do I need to test?



MODULE 64 | APPLY AND KEEP

Turn the idea into a professional record.

APPLY

Where does this appear in my organization? Use one real task, decision, record, role, asset, or consequence.
Learning job: Present the case, answer challenges, revise the plan, and state the next accountable action

KEEP

Add this record to the implementation portfolio: defended implementation portfolio.

EVIDENCE

What source or observation supports this?

OWNER

Who remains accountable?

NEXT ACTION

What will I test, ask, change, or stop?

COMPLETION CHECK

☐ lesson ☐ activity ☐ record ☐ reflection ☐ ready for review



CAPSTONE PORTFOLIO

Assemble the record. Do not hide the unknowns.

[] 01	Executive summary and utility problem
[] 02	Current process and baseline
[] 03	Opportunity portfolio and selected use case
[] 04	Data, knowledge, sources, and limitations
[] 05	Architecture decision and alternatives
[] 06	Governance, security, permissions, and human authority
[] 07	Evaluation, economics, and success measures
[] 08	Workforce and operating-model implications
[] 09	Ninety-day pilot brief
[] 10	One-year roadmap
[] 11	Unresolved questions and next accountable decision



SIX MONTHS LATER

What can I now explain, decide, build, and stop?

RECOGNITION

What do I see now that I did not see at the beginning?

PROFESSIONAL JUDGMENT

Which decision can I make more clearly, and what evidence will I require?

NEXT ACCOUNTABLE ACTION

What happens in the next three hours, the next 30 days, and the next 90 days?
